Do Vision-Language Models Really See What They Say?

A Comprehensive Evaluation of Hallucinations in Large Vision-Language Models

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Abstract

Vision-Language Models (VLMs) have demonstrated remarkable capabilities in understanding and describing images, yet they remain prone to hallucinations (confident descriptions of visual content not actually present in the image). This report evaluates hallucination patterns across four critical categories: absent objects, incorrect attributes, spatial relations, and counting errors. I developed a synthetic probing benchmark to systematically test two state-of-the-art VLMs: LLaVA-v1.5 (7B) and Qwen3-VL (8B-Instruct). Results reveal significant differences in hallucination susceptibility: LLaVA-v1.5 exhibits a 47% false-positive rate with particular vulnerability in spatial reasoning tasks, while Qwen3-VL demonstrates superior robustness with only an 8% false-positive rate across all categories. My findings underscore the importance of careful model selection for safety-critical applications and provide a foundation for developing more reliable vision-language systems.

1 Introduction

Recent advances and research in Vision-Language Models (VLMs) have enabled machines to describe images with fluency and semantic understanding. Nowadays, models such as GPT-4 Vision, LLaVA, and Qwen-VL have achieved impressive performance on numerous downstream tasks, from visual question answering to image captioning [3, 4]. However, a critical limitation persists: these models frequently hallucinate content that is not present in the input image.

Hallucination in VLMs refers to the generation of confident descriptions about visual elements, attributes, or relationships that have no basis in the actual image. This phenomenon causes significant risks for real-world deployment:

- **Safety-Critical Applications**: In medical imaging analysis, autonomous driving systems, and automated document processing, hallucinated visual content could lead to dangerous or unfixable errors.
- **Reliability**: Users and downstream systems may rely on model outputs without verification, assuming accuracy proportional to confidence scores.

Understanding hallucination patterns is essential for developing more reliable VLMs. Previous work has documented hallucination phenomena in large language models and vision-language systems [1, 2], but systematic comparative evaluation across multiple hallucination types remains limited.

This report presents a comprehensive evaluation of hallucination in two prominent open-source VLMs using a carefully designed synthetic benchmark. My contributions in this project are:

1. A systematic evaluation framework targeting four distinct hallucination categories.
2. Quantitative comparison of LLaVA-v1.5 and Qwen3-VL across hallucination metrics.
3. Detailed analysis of which visual conditions and linguistic contexts make hallucinations more likely.
4. Comprehensive mitigation strategy development and analysis through prompt engineering evaluation.
5. Evidence-based recommendations for practitioners selecting VLMs for sensitive applications.

2 Related Work

Hallucination in large language models manifests as convincing but ultimately inaccurate text generation. In vision-language systems, hallucination compounds this by integrating errors across visual and linguistic modalities.

Li et al. [2] introduced the POPE (Poll-based Object Probing Evaluation) framework for measuring object hallucination rates. My evaluation extends POPE-style metrics to four hallucination categories: absent objects, attribute errors, spatial relations, and counting. Specifically, I use POPE negative, where the correct answer to all probes is "no", making this a negative probing benchmark designed to measure hallucination rates through false-positive detection.

LLaVA-v1.5 [3] combines CLIP vision encoders with Vicuna-7B language model, while Qwen3-VL [4] represents the latest generation with improved vision-language alignment. Despite architectural advances, hallucination remains a critical limitation for deployment even today.

3 Methodology

3.1 Benchmark and Models

I have constructed a synthetic benchmark of 100 image-question pairs across four hallucination categories: (1) Absent Objects (25), (2) Attribute Errors (25), (3) Spatial Relations (25), (4) Counting (25). I evaluated two open-source models in default configurations without fine-tuning: LLaVA-v1.5 (7B, CLIP + Vicuna) and Qwen3-VL (8B-Instruct, improved vision-language alignment). In addition, I present the effectiveness of a mitigation strategy applied to each model to reduce hallucination rates.

3.2 Metrics

Overall Accuracy on trap questions, Hallucination Rate (false-positive rate), Category-specific performance, and Prompt engineering effects (baseline vs. strict prompting).

3.3 Benchmark Examples

To illustrate the evaluation methodology, I present concrete examples from each hallucination category:

Examples:

- **object_absent**: Image contains a blue square. Question: "Is there a red dog?" Correct answer: "No".
  - If model says "No" $\Rightarrow$ TN
  - If model says "Yes" $\Rightarrow$ FP (hallucinated a non-existent object)
- **attribute**: Image contains a blue square. Question: "Is the square red?" Correct answer: "No".
  - If model says "No" $\Rightarrow$ TN
  - If model says "Yes" $\Rightarrow$ FP (hallucinated an incorrect attribute)
- **relation**: Circle positioned below triangle. Question: "Is the circle above the triangle?" Correct answer: "No".
  - If model says "No" $\Rightarrow$ TN
  - If model says "Yes" $\Rightarrow$ FP (hallucinated an incorrect spatial relation)
- **count**: Image contains 3 circles. Question: "Are there 5 circles?" Correct answer: "No".
  - If model says "No" $\Rightarrow$ TN
  - If model says "Yes" $\Rightarrow$ FP (hallucinated an incorrect count)

4 Results

4.1 Overall Performance

My evaluation reveals stark differences in hallucination susceptibility between the two models:

Metric	LLaVA-v1.5	Qwen3-VL
Overall Accuracy	53%	92%
Hallucination Rate (FPR)	47%	8%
Total Samples	100	100

Table 1: Overall Hallucination Metrics Comparison

This substantial performance gap (47 percentage points in hallucination rate) indicates fundamental differences in how these models process and integrate visual information with question context.

4.2 Category-Specific Analysis

4.2.1 Absent Objects

Both models perform well on this task, with Qwen3-VL achieving perfect accuracy. LLaVA-v1.5's 76% accuracy suggests occasional false-positive errors when explicitly asked about absent objects. This is the category where LLaVA-v1.5 performs relatively best, achieving nearly three-quarters correct identifications of missing objects.

Model	**Accuracy**	**Samples**
LLaVA-v1.5	76%	25
Qwen3-VL	100%	25

Table 2: Absent Object Recognition Performance

4.2.2 Attribute Errors

Attribute recognition shows moderate robustness in LLaVA-v1.5 (68%) but near-perfect performance in Qwen3-VL (96%). This suggests that attribute hallucination—incorrectly describing object colors, materials, or other properties—is a notable weakness in older VLM architectures.

Model	**Accuracy**	**Samples**
LLaVA-v1.5	68%	25
Qwen3-VL	96%	25

Table 3: Attribute Hallucination Analysis

4.2.3 Spatial Relations (Critical Weakness)

This result represents LLaVA-v1.5's most critical weakness: spatial reasoning. With only 12% accuracy on spatial relationship questions, the model hallucinates spatial configurations in 88% of cases. Qwen3-VL performs substantially better (80%), though this remains the weakest category for both models, indicating that precise spatial reasoning remains a challenge for current VLMs.

Model	**Accuracy**	**Samples**
LLaVA-v1.5	12%	25
Qwen3-VL	80%	25

Table 4: Spatial Reasoning Performance

4.2.4 Counting

LLaVA-v1.5 shows near-random performance on counting tasks (56%), indicating significant difficulty with numerical quantification of visual content. Qwen3-VL demonstrates much better performance (92%), suggesting improved numerical reasoning capabilities in newer architectures.

Model	**Accuracy**	**Samples**
LLaVA-v1.5	56%	25
Qwen3-VL	92%	25

Table 5: Counting Task Performance

4.3 Comparative Visualization

Category	LLaVA (Acc.)	Qwen (Acc.)	Delta	Gap Analysis
Absent Objects	76%	100%	+24 pp	Minor weakness
Attributes	68%	96%	+28 pp	Moderate weakness
Spatial Relations	12%	80%	+68 pp	Critical Gap
Counting	56%	92%	+36 pp	Significant weakness
Overall	**53%**	**92%**	**+39 pp**	

Table 6: Comprehensive Category Breakdown

4.4 Mitigation Strategy: Prompt Engineering

As I commented before, I evaluated prompt engineering as a low-cost mitigation approach by comparing:

Baseline: "[Question]? Answer only yes or no."

Strict: "[Question]? Answer only yes or no. Verify the visual evidence carefully. If the requested object, attribute, relation, or count is not clearly visible, answer no. Do not infer likely objects from context."

4.4.1 Results

Strict prompting reduces hallucinations for LLaVA-v1.5 (+10 pp accuracy, -10 pp hallucination), with the largest improvements in spatial reasoning (+12 pp) and counting (+32 pp). Qwen3-VL shows different behavior: accuracy slightly decreases (-1 pp), hallucination slightly increases (+1 pp), and spatial reasoning decreases (-4 pp). This suggests that for already-robust models, strict prompting introduces unnecessary conservatism, while for weak models, it provides substantial benefits, being architectural improvements the most effective hallucination mitigation strategy.

	LLaVA-v1.5		Qwen3-VL	
Metric	Base	Strict	Base	Strict
Accuracy	53%	63%	92%	91%
Hallucination	47%	37%	8%	9%
Spatial	12%	24%	80%	76%

Table 7: Mitigation Study: Baseline vs Strict Prompting

4.4.2 Category-Specific Improvements

For LLaVA-v1.5, counting shows the largest improvement (+32 pp), suggesting the model benefits significantly from explicit verification instructions when quantifying objects. Spatial reasoning also improves substantially (+12 pp), though it remains weak. Attributes show minor degradation, indicating strict prompting may be overly conservative for attribute verification.

Category	Baseline	Strict	Improvement
Absent Objects	76%	76%	0 pp
Attributes	68%	64%	-4 pp
Spatial Relations	12%	24%	+12 pp
Counting	56%	88%	+32 pp

Table 8: LLaVA-v1.5: Category Performance with Mitigation

For Qwen3-VL, strict prompting provides no improvements and causes minor regression in spatial reasoning (-4 pp). This indicates that the model already incorporates evidence-based reasoning implicitly, and explicit instructions actually introduce unnecessary constraints.

Category	Baseline	Strict	Change
Absent Objects	100%	100%	0 pp
Attributes	96%	96%	0 pp
Spatial Relations	80%	76%	-4 pp
Counting	92%	92%	0 pp

Table 9: Qwen3-VL: Category Performance with Mitigation

5 Discussion

5.1 Key Findings

My evaluation reveals that while Vision-Language Models have advanced substantially, hallucination remains a critical limitation. LLaVA-v1.5's catastrophic failure on spatial relations (12% accuracy) suggests weak spatial grounding. Qwen3-VL demonstrates improved visual-semantic alignment (92% overall, 80% spatial), indicating that recent architectural improvements substantially reduce hallucination. This 39-percentage-point gap underscores the importance of model selection for safety-critical applications.

5.2 Recommendations

For safety-critical applications, Qwen3-VL is substantially preferable to LLaVA-v1.5 across all hallucination dimensions. However, no single VLM should be trusted alone for spatial reasoning tasks. I recommend: (1) model selection based on task-specific requirements, (2) strict prompting to reduce false-positives, (3) ensemble approaches for critical decisions, and (4) human verification for high-stakes applications.

6 Conclusion

This report presents a systematic evaluation of hallucination susceptibility in two open-source Vision-Language Models using a controlled negative-probing benchmark. The results reveal a substantial performance gap: LLaVA-v1.5 achieves only 53% accuracy with a 47% hallucination rate, while Qwen3-VL reaches 92% accuracy with an 8% false-positive rate, a 39-percentage-point difference that reflects meaningful architectural progress in vision-language alignment.

Category-level analysis identifies spatial reasoning as the most persistent challenge across both models. LLaVA-v1.5 fails near-catastrophically in this category (12%), and even Qwen3-VL, despite its overall robustness, achieves only 80%, suggesting that geometric and relational understanding remains an open problem for current VLMs regardless of scale or training improvements.

Prompt engineering offers a practical, zero-cost mitigation for weaker models: strict verification instructions improve LLaVA-v1.5's overall accuracy by 10 percentage points, with particularly large gains in counting (+32 pp) and spatial reasoning (+12 pp). However, the same strategy yields no benefit—and minor regression—for Qwen3-VL, indicating that explicit prompting constraints become counterproductive once a model has internalized evidence-based reasoning during training.

Taken together, these findings carry clear practical implications. Model selection is the single most impactful decision for reducing hallucination risk: for any application where false positives carry meaningful consequences, architectural quality dominates prompt-level interventions. Nevertheless, no current model should be deployed without human oversight in spatial or safety-critical tasks. Future work should extend this evaluation to larger benchmarks with natural images, multi-object scenes, and adversarial linguistic framings to better characterize the boundaries of hallucination robustness in production settings.

References

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